Track 9C

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Contents

Track 9C — Frontier Domain Exploration: Model Generalizability Test **2**
Purpose . . . 2
Approach: Prediction-Verification Paradigm . . . 2
Candidate Domains . . . 2
Methodology . . . 3
Deliverables . . . 3
Convergence Criteria . . . 3

Track 9C — Frontier Domain Exploration: Model Generalizability Test



Figure 1: Track 9C Frontier Domain Exploration — Plan Overview Infographic

Purpose

Apply the model to domains outside the original 30 to test the generalizability of structural themes. This is the most direct test of whether the model can predict, not just explain after the fact.

Approach: Prediction-Verification Paradigm

Before investigating a new domain, we predict how each structural theme should manifest. Then we verify through actual research. The hit rate becomes a measure of the model's predictive power.

Candidate Domains

Domain	Selection Rationale	Prediction Difficulty
Consciousness / Neuroscience	Boundary concepts may be fundamentally different	High
Urban Planning / Architecture	Physical instantiation of "edges"	Medium
Game Theory	Formally defined structures; easier to compare	Medium

Domain	Selection Rationale	Prediction Difficulty
Quantum Information Theory	Stress test for cross-scale self-similarity	High
Culinary Arts / Food Culture	Most alien to the existing 30 domains	High
Law / Jurisprudence	Institutional boundaries with unique typology	Medium

Methodology

Step 1: Domain Selection

Select one domain from candidates. Start with medium-difficulty predictions.

Step 2: Blind Prediction

Before consulting any domain-specific literature, predict how each of the five structural themes would manifest. Assign confidence levels (High / Medium / Low) and record predictions with timestamps.

Step 3: Domain Investigation

Research the creation and generative processes within the selected domain. Collect findings on boundaries, transitions, circulation, and scale structures.

Step 4: Prediction Audit

Open the blind predictions and compare against investigation results. Judge each as hit, partial hit, or miss. For misses, analyze whether the failure reflects model limitations or prediction methodology issues.

Step 5: Model Update Proposal

- Hits: Record as evidence of model generalizability
- Misses: Propose additional conditions or new theme candidates

Deliverables

For each domain:

- **Blind predictions** — Timestamped pre-investigation predictions
- **Domain survey** — Investigation results on creative structures
- **Prediction audit** — Hit rate analysis and assessment
- **Model update proposal** — Generalizability evidence or identified limitations

Convergence Criteria

- Three or more domains have completed the prediction-verification cycle
- Hit rate trends are visible (improving, stable, or declining)
- Model applicability boundaries have been concretely identified